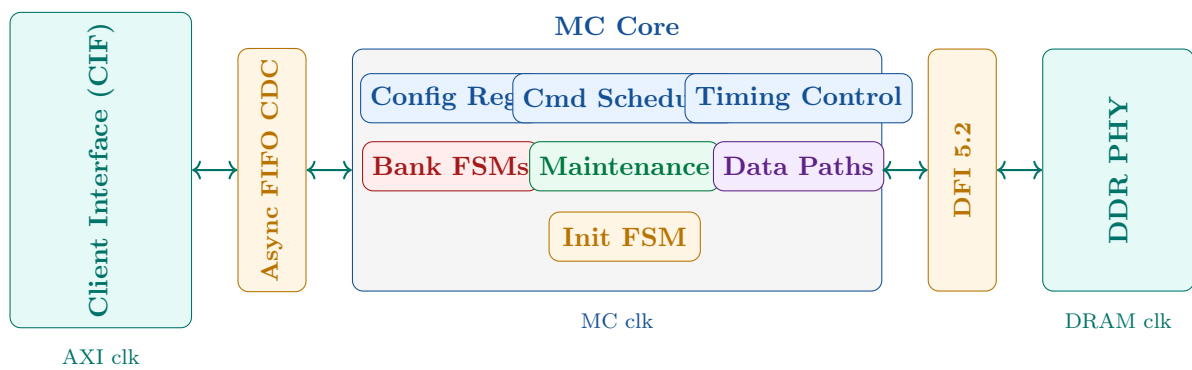


MC Core

Design Specification

Reconfigurable DDR5 Memory Controller



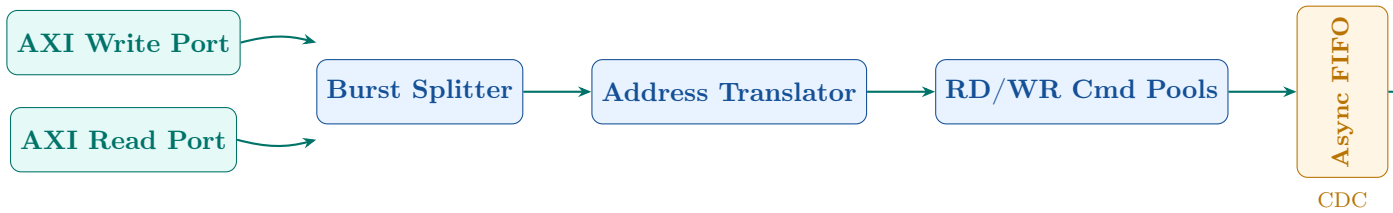
Standard	JEDEC JESD79-5 (DDR5), DFI 5.2, AXI4
Banks	16 (2 BG × 4 banks, per sub-channel)
Burst	BL16 default (BL/2 = 8 clock cycles)
Clock	Dual domain: AXI (CIF) / MC (DRAM), CDC at one crossing point
FSMs	23 total: 1 Init + 16 BSM + 4 Maintenance + Scheduler + CRC + ECS
DFI freq	1:1, 1:2, 1:4 ratio supported

Contents

System Overview

The MC Core is the central engine of the Reconfigurable Memory Controller (RMC). It sits entirely in the MC clock domain, receiving commands from the Client Interface (CIF) via a single Async FIFO CDC bar, and emitting DFI 5.2 commands toward the DDR PHY. Every DRAM timing constraint, scheduling decision, refresh housekeeping, and power-state transition is the MC Core's responsibility.

Top-Level Signal Flow



Sub-Block Summary

Sub-Block	Domain	Role
Config Registers	MC	Holds all timing knobs, page policy, refresh mode, RFM thresholds, DFI latency offsets. Written via AXI4-Lite CSR path.
Init FSM	MC	Full DDR5 power-up sequence: RESET_n, NOP burst, DLL reset, ZQcal via MPC, MRW burst, optional training dispatch. Single instance.
Bank State Machines	MC	16 instances. Each tracks its bank's state and owns all per-bank timers. Source of truth for bank availability.
Timing Control	MC	Combinational + registered timer array. Produces <code>cmd_ok[15:0]</code> <code>[cmd_type]</code> each cycle for the Scheduler.
Arbitration Logic	MC	Picks the winner command class each MC clock cycle. Implements priority, age-based anti-starvation, and watermark-controlled RD/WR mode switching.
Command Selector	MC	Takes the winner class, applies all DRAM timing constraints, enforces page policy, emits the final command tuple {cmd, rank, BG, bank, row, col}.
Refresh Scheduler	MC	Leaky-bucket tREFI tracking. REFab and REFsb modes. Handles up to 8× postponement; asserts <code>ref_urgent</code> on overflow.
ZQcal FSM	MC	Periodic MPC ZQCAL Start/Latch sequence.
RFM FSM	MC	DDR5 rowhammer mitigation. Per-bank RAA counters; triggers RFM command when $RAA \leq RAA_{IMT}$.
Power Management FSM	MC	CKE-controlled transitions: Active PD, Precharge PD, Self-Refresh, MPSM.
Write Data Path	MC	BL16-aligned write buffer, DM/ECC generation, Write CRC (CRC-8 over DQ+DM), DFI write timing pipeline.

Read Data Path	MC	Read data capture FIFO, Read CRC/ECC checker, read latency counter, ECS FSM.
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Config Registers

All timing parameters consumed by the MC Core live here. The host programs them before asserting `INIT_KICK` and may update them at runtime only while the controller is idle. The scheduler reads timing values combinatorially every cycle.

Register	Width	Description
CL	7b	CAS Latency in nCK. Drives RL used in RTW formula.
CWL	7b	CAS Write Latency = CL-2 (fixed in DDR5).
tRCD	8b	RAS-to-CAS delay (nCK). ACT to first RD/WR on same bank.
tRP	8b	Row Precharge Time (nCK). PRE to next ACT on same bank.
tRAS	9b	Row Active Time minimum (nCK). ACT to PRE minimum.
tRC	9b	Row Cycle Time = tRAS+tRP. ACT to ACT same bank.
tWR	8b	Write Recovery Time (nCK). WR burst end to PRE.
tRTP	6b	Read-to-Precharge (nCK). RD cmd to PRE on same bank.
tCCD_L	5b	CAS-to-CAS, same BG (nCK).
tCCD_S	5b	CAS-to-CAS, diff BG = BL/2 = 8 nCK.
tCCD_L_WR	6b	WR-to-WR, same BG (nCK). max(32 nCK, 20 ns) per spec.
tCCD_L_WR2	6b	WR-to-WR, same BG, 2nd not RMW (nCK). max(16 nCK, 10 ns).
tWTR_L	6b	Write-to-Read same BG (nCK).
tWTR_S	5b	Write-to-Read diff BG (nCK).
tRRD_L	5b	ACT-to-ACT same BG (nCK).
tRRD_S	4b	ACT-to-ACT diff BG (nCK).
tFAW	6b	Four-Activate Window (nCK).
tRFC1	12b	REFab recovery time (nCK). 195 ns for 8 Gb DDR5.
tRFCsb	11b	REFsb per-bank recovery (nCK).
tREFI	14b	Average refresh interval (nCK). 3.9 μ s at $\leq 85^\circ\text{C}$.
tMRD	5b	MRW-to-MRW minimum (nCK). 16 nCK in DDR5.
tXP	5b	PDX-to-command (nCK).
tXS	12b	SRX-to-command without DLL lock (nCK). tRFC1+10 ns.
tCKSRE	5b	Stable clock cycles required after SRE.
tCKSRX	5b	Stable clock cycles required before SRX.
tDLLK	11b	DLL lock time after DLL Reset MPC. 1024 nCK.
RAAIMT	8b	RFM RAA Initial Management Threshold (from MR58).
RAAMMT	8b	RFM RAA Maximum Management Threshold (from MR58).
RAADec	4b	RAA decrement per REF command issued.
REF_MODE	2b	00=REFab, 01=REFsb, 10=FGR-2x, 11=FGR-4x.

Register	Width	Description
PAGE_POLICY	2b	00=Open, 01=Closed, 10=Adaptive.
AGE_THR1	8b	Intra-class age escalation threshold (cycles).
AGE_THR2	8b	Cross-class age escalation threshold (cycles).
WR_HIGH_WM	6b	Write pool watermark high — triggers switch to WR mode.
WR_LOW_WM	6b	Write pool watermark low — triggers switch back to RD mode.
PHY_WRLAT	6b	<code>dfi_t_phy_wrlat</code> : WL pipeline offset to PHY.
PHY_RDLAT	6b	<code>dfi_t_rddata_en</code> : RL pipeline offset to PHY.
FREQ_RATIO	2b	DFI frequency ratio: 00=1:1, 01=1:2, 10=1:4.
INIT_KICK	1b	Write 1 to trigger Init FSM.
SOFT_RESET	1b	Synchronous soft-reset of all MC Core FSMs.

Init FSM

Reference: JESD79-5 §3.3.1 (power-up with stable power) and §3.3.2 (reset with stable power). Single instance. Triggered by `INIT_KICK`. Emits DFI commands directly, bypassing the Cmd Scheduler. Releases `init_done` on completion.

States and Transitions

State	Action	Exit Condition
IDLE	Hold RESET_n=0, CKE=0, CS_n=0. All DFI outputs deasserted.	<code>INIT_KICK</code> =1
ASSERT_RESET	RESET_n held LOW. CS_n held LOW. Load tINIT1 (200 µs) counter.	tINIT1 timer expires
CS_LOW_PRE_DEASSERT	CS_n held LOW. Count tINIT2 (10 ns) before releasing RESET_n.	tINIT2 timer expires
CS_LOW_POST_DEASSERT	Deassert RESET_n (HIGH). CS_n remains LOW. CKE still LOW. Count tINIT3 (4 ms).	tINIT3 timer expires
ODT_SETTLE	Raise CKE. Stable CK already running. Count tINIT4 (2 µs) for ODT and CK settle.	tINIT4 timer expires
NOP_BURST	Issue ≥3 NOP/DES on DFI. CS_n driven normally.	tINIT5 NOP count satisfied
WAIT_XPR	Count tXPR = max(5 nCK, tRFC1+10 ns) after CS_n first HIGH.	tXPR timer expires
MPC_DLL_DIVIDER	Issue MPC opcode: DLL Divider Select. Configures tCCD_L divider via MR13.	DFI command accepted
MPC_DLL_RESET	Issue MPC opcode: DLL Reset. Required before ZQcal per spec.	DFI command accepted

State	Action	Exit
ZQCAL_START	Issue MPC opcode: ZQCAL Start (OP=0x00). Initiates impedance calibration.	DFI command accepted
WAIT_tZQCAL	Wait. DRAM performs internal ZQ calibration.	tZQCAL (1 μ s) expires
ZQCAL_LATCH	Issue MPC opcode: ZQCAL Latch (OP=0x01). Captures calibrated driver/ODT codes.	DFI command accepted
WAIT_tZQLAT	Wait tZQLAT = 30 nCK for latch propagation.	tZQLAT timer expires
MRW_BURST	Issue MRW to all required mode registers in order: MR0 (BL, CL), MR2 (functional modes), MR4 (refresh), MR5 (ODT/DM), MR6 (tWR/tRTP), MR8 (burst/CRC/preamble), MR32–MR35 (ODT), MR58 (RFM). One MRW per tMRD (16 nCK).	All required MRWs issued
TRAINING_DISPATCH	Optional. Controlled by TRAIN_EN register bits. Dispatches CA training, CS training, Write Leveling (MR2 OP[5]=1), and read training (MPC RTP Start) sub-sequences via DFI training handshake signals.	All enabled training sub-sequences complete, or skipped if TRAIN_EN =0
DONE	Assert init_done . Release Cmd Scheduler from reset. Init FSM idles until soft-reset or re-kick.	Permanent until SOFT_RESET

MRW Sequence Detail

The MRW burst programs the following registers in order. Each write is separated by tMRD (16 nCK).

Register	Fields Programmed
MR0	BL = BL16 (OP[1:0]=00), CL from CL config reg (OP[6:2])
MR2	2N mode, MPSM, write leveling enable if TRAIN_EN [WL]=1
MR4	Refresh rate per temperature; FGR mode if REF_MODE [1]=1
MR5	RTT PARK, DM enable
MR6	tWR and tRTP from tWR , tRTP config regs
MR8	Burst length, preamble mode (2/4 tCK), Write CRC enable, Read CRC enable
MR32	CK ODT, CS ODT
MR33	CA ODT, DQS RTT PARK
MR34	RTT PARK, RTT WR
MR35	RTT NOM WR, RTT NOM RD

Reg	Fields
MR58	Read-only (RAAIMT, RAAMMT). Init FSM reads via MRR and mirrors to RAAIMT / RAAMMT config regs.

Bank State Machines (BSM)

16 instances, one per bank (2 BG $\times$ 4 banks $\times$ 2 sub-channels per device, or 4 BG $\times$ 4 banks for x4/x8 devices). Each BSM is independent. It receives commands from the Cmd Scheduler and `cmd_ok` enables from Timing Control. All per-bank timers reside inside the BSM.

States and Transitions

State	Legal Input Commands	Entry / Timer / Exit
IDLE	ACT, REFab, REFsb, PDE, SRE, RFM	Default state. No timer running. Entered from PRECHARGING when tRP expires, from REFRESHING when tRFC expires, or at init.
ACTIVATING	None (internal delay)	Entered on ACT command. Runs tRCD countdown. Transitions to BANK_ACTIVE when tRCD expires.
BANK_ACTIVE	RD, RDA, WR, WRA, PRE, PREA	Entered from ACTIVATING. tRAS timer loaded at entry. PRE only issued after tRAS satisfied. Open row address registered for row-hit detection.
READING	BST (truncate to BC8)	Entered on RD or RDA command. Bus occupied for CL+BL/2 cycles. If RDA: hidden precharge starts internally at tRTP after RD. Exits to BANK_ACTIVE (no AP) or PRECHARGING (RDA).
WRITING	BST (truncate to BC8)	Entered on WR or WRA command. Bus occupied for CWL+BL/2 cycles. If WRA: hidden precharge starts at CWL+BL/2. Exits to BANK_ACTIVE (no AP) or PRECHARGING (WRA).
PRECHARGING	None (internal delay)	Entered on PRE/PREA or at end of RDA/WRA burst. tRP counter loaded. Transitions to IDLE when tRP expires.
REFRESHING_AB	None	Entered when Refresh Scheduler issues REFab. All 16 banks enter this state simultaneously. tRFC1 counter loaded. All banks return to IDLE when tRFC1 expires. FGR mode: tRFC2 or tRFC4 used instead.
REFRESHING_SB	None	Entered when Refresh Scheduler issues REFsb targeting this bank. Only the targeted bank is locked; other banks remain accessible. tRFCsb counter loaded. Exits to IDLE when tRFCsb expires.

State	Legal Inputs	Entry/Timer/Exit
POWER_DOWN	PDX	Entered from IDLE (Precharge PD) or BANK_ACTIVE (Active PD) when CKE deasserted. tCKE minimum low time enforced. Exits to IDLE or BANK_ACTIVE respectively after PDX + tXP.
SELF_REFRESH	SRX	Entered from IDLE (all banks must be precharged first). tCKSRE stable clock enforced after entry. Minimum tCKESR in SR. On SRX: tCKSRX stable clock required, then tXS before any command.
RFM_ACTIVE	None	Entered when RFM FSM issues an RFM command to this bank ($RAA \leq RAA_{IMT}$). tRFM counter loaded (same duration as tRFC1). Exits to IDLE when tRFM expires.

Per-Bank Timer Table

Timer	Loaded On	Guards
tRCD counter	ACT issued	First RD/WR on same bank
tRAS counter	ACT issued	PRE/PREA on same bank (minimum open)
tRC counter	ACT issued	Next ACT on same bank
tRP counter	PRE issued	Next ACT on same bank
tWR counter	WR burst end	PRE on same bank
tRTP counter	RD cmd issued	PRE on same bank
tRFC1 counter	REFab issued	Any command on any bank (all banks blocked)
tRFCsb counter	REFsb issued	Commands to the refreshed bank only
tXS counter	SRX asserted	First ACT after self-refresh exit
tXP counter	PDX asserted	First command after power-down exit
tMRD counter	MRW issued	Next MRW command

Timing Control

The Timing Control block is not a state machine. It is a **combinational and registered timer array** that produces `cmd_ok[15:0]` [[CMD_TYPES](#)] every MC clock cycle. The Cmd Scheduler may only issue a command if the corresponding `cmd_ok` bit is asserted.

Per-Bank Countdown Timers

Each of the 11 per-bank timers (listed above in §4.2) is implemented as a down-counter. When the counter reaches zero the relevant `cmd_ok` bit is set. The timers are loaded in the same cycle the triggering command is issued to DFI.

Cross-Bank Constraint Checkers

Constraint	Scope	Implementation
tRRD_L	Same BG, any bank	Timestamp of last ACT in same BG stored. New ACT blocked until $T_{\text{now}} - T_{\text{last ACT in BG}} \geq tRRD_L$.
tRRD_S	Diff BG, any bank	Timestamp of last ACT globally. New ACT in a different BG blocked until $T_{\text{now}} - T_{\text{last ACT any}} \geq tRRD_S$.
tFAW	All banks, same rank	4-entry ring buffer of the most recent ACT timestamps. New ACT blocked if the oldest of the 4 timestamps is within tFAW cycles. $\text{faw_ok} = (\text{now} - \text{act_ts}[3]) \geq tFAW$.
tCCD_L	Same BG, RD/WR	Timestamp of last CAS in same BG. Next CAS in same BG blocked until $\Delta T \geq tCCD_L$. For WR-to-WR same BG, tCCD_L_WR is used. For WR-to-WR same BG when 2nd write is not RMW: tCCD_L_WR2.
tCCD_S	Diff BG, RD/WR	Timestamp of last CAS globally. Next CAS in different BG blocked until $\Delta T \geq tCCD_S$ ($= BL/2 = 8 nCK$ in DDR5).
tWTR_L	Same BG, WR→RD	Timestamp of last write data end in same BG. RD command blocked until $\Delta T \geq WL + BL/2 + tWTR_L$.
tWTR_S	Diff BG, WR→RD	Same but using tWTR_S. Enforces $\Delta T \geq WL + BL/2 + tWTR_S$.
tRTW	Any, RD→WR	Timestamp of last RD command. WR command blocked until $\Delta T \geq RL + BL/2 - WL + 2 + tWPRE$. ($tWPRE = 2 tCK$ default per MR8.)

Arbitration Logic

Runs every MC clock. Answers: *which command class wins this cycle?* Does not select bank, row, or column — that is the Command Selector’s job. Outputs: `arb_winner[2:0]`, `pool_ptr`, `mode_state`, `ref_urgent`.

Priority Hierarchy

Level	Winner	Condition	Effect
0	REF urgent	REF credits ≥ 8 , or time since last REF $> tREFI_{\text{max}}$	Hard stall — all RD/WR suppressed
1	REF nominal	tREFI countdown reaches zero	REF injected; RD/WR deferred
2	ZQ calibration	tZQCS interval expires	ZQCS after current REF completes
3	Read	RD pool non-empty AND RD mode active	Promoted; BG-aware scheduling
4	Write	WR pool non-empty AND WR mode active	Batched; watermark-controlled

Level	Winner	Condition	Effect
5	Scrubber / MRR	Background maintenance, idle slot	Lowest; pre-emptible

Age Counter Logic (Anti-Starvation)

Every RD/WR pool entry carries an age counter incremented each MC cycle it sits unserved.

- Age counter $\geq$ **AGE_THR1** (default 64 cycles): entry jumps to head of its class (intra-class HOL bypass).
- Age counter $\geq$ **AGE_THR2** (default 256 cycles): WR entry may overtake the RD priority level (cross-class escalation, configurable).
- Age counter reset to 0 on dispatch.

RD/WR Mode Switching (Watermark Logic)

Prevents excessive tWTR/tRTW turnaround penalties by batching reads and writes into phases.

- **Switch to WR mode:** WR pool depth $\geq$ **WR_HIGH_WM** (default 32 entries), OR RD pool empty and WR pool non-empty.
- **Switch back to RD mode:** WR pool depth $\leq$ **WR_LOW_WM** (default 8 entries), OR WR pool empty and RD pool non-empty, OR urgent RD due to age escalation.
- Hysteresis gap ($HIGH - LOW = 24$ entries) prevents mode thrashing.

REF Credit Tracker

- REF credits $+= 1$ every tREFI.
- REF credits $-= 1$ on each REF issued.
- **ref_urgent** = (credits ≥ 8) OR ($\Delta t_{\text{since last REF}} > tREFI_{\text{max}}$).
- JEDEC allows a maximum of 8 postponed refreshes. When **ref_urgent** asserts, arbitration outputs REF unconditionally regardless of pool state.
- Temperature derating: at $>85^{\circ}\text{C}$ (detected from MR4 TUF bit), tREFI is halved by the Refresh Scheduler before loading into the credit tracker.

Command Selector

Takes the winner class from Arbitration Logic, applies all remaining DRAM timing constraints, and emits the fully resolved command: {cmd_type, rank, BG, bank, row, col, ap, wr_partial}.

Page Policy Engine

Policy	Row Hit	Row Miss / Empty
Open Page (default)	CAS only	Miss: $\text{PRE} \xrightarrow{t_{RP}} \text{ACT} \xrightarrow{t_{RCD}} \text{CAS}$. Empty: $\text{ACT} \xrightarrow{t_{RCD}} \text{CAS}$.
Closed Page	ACT + CAS (always)	$\text{ACT} \xrightarrow{t_{RCD}} \text{CAS}$. Best for random/streaming workloads.

Policy	Hit	Miss/Empty
Adaptive	Starts as Open	Hit-rate tracked over sliding window $N = 256$ accesses. Hit rate > OPEN_THR (70%): stay open. Hit rate < CLOSE_THR (30%): switch to closed. Autotunes per bank independently.

Bank-Group Aware Reordering

DDR5 imposes a large same-BG write penalty: $t_{CCD_L_WR} = \max(32 \text{ nCK}, 20 \text{ ns})$ vs. $t_{CCD_S} = 8 \text{ nCK}$ for different BG. The Command Selector scans the candidate queue for cross-BG entries to exploit the shorter gap. Reordering is bounded by the age limit to prevent starvation.

Transition	Same BG	Diff BG
RD $\rightarrow$ RD	$t_{CCD_L} = \max(8 \text{ nCK}, 5 \text{ ns})$	$t_{CCD_S} = 8 \text{ nCK}$
WR $\rightarrow$ WR	$t_{CCD_L_WR} = \max(32 \text{ nCK}, 20 \text{ ns})$	$t_{CCD_S} = 8 \text{ nCK}$
WR $\rightarrow$ WR (2nd not RMW)	$t_{CCD_L_WR2} = \max(16 \text{ nCK}, 10 \text{ ns})$	t_{CCD_S}

Legal Check Matrix

Before dispatching any candidate command, all of the following gates must be true simultaneously. A false on any gate causes a NOP to be inserted that cycle.

Gate	Timer Source	Constraint
bank_avail[b]	BSM state	Bank not in ACTIVATING or PRECHARGING
gate_rcd[b]	ACT issue	t_{RCD} : ACT to first CAS on same bank
gate_rp[b]	PRE issue	t_{RP} : PRE to ACT on same bank
gate_ras[b]	ACT issue	t_{RAS} : ACT to PRE minimum
gate_ccd_l	Same BG CAS	t_{CCD_L} or $t_{CCD_L_WR}$ or $t_{CCD_L_WR2}$
gate_ccd_s	Diff BG CAS	$t_{CCD_S} = BL/2$
gate_rrd_l	Same BG ACT	t_{RRD_L}
gate_rrd_s	Diff BG ACT	t_{RRD_S}
gate_faw	Sliding window	≤ 4 ACTs within t_{FAW}
gate_wtr_l	WR data end, same BG	t_{WTR_L} : WR to RD
gate_wtr_s	WR data end, diff BG	t_{WTR_S} : WR to RD
gate_rtw	RD cmd	$t_{RTW} = RL + BL/2 + 2 - WL$
gate_rfc[r]	REF issue (per rank)	t_{RFC1} : all commands blocked
gate_rfcpb[r][b]	REFsb (per bank)	t_{RFCsb} : same bank only blocked
gate_zqcs	ZQCS issue	All commands blocked during ZQ
gate_mrd	MRW issue	t_{MRD} : MRW to MRW

Command Sequence Generator

Outputs into a 4-deep command FIFO. During ACT/PRE wait cycles, other banks may be serviced (pipelining).

Page Decision	Emitted Sequence
Row Hit	CAS (RD/WR)
Row Miss	PRE $\xrightarrow{t_{RP}}$ ACT $\xrightarrow{t_{RCD}}$ CAS
Row Empty	ACT $\xrightarrow{t_{RCD}}$ CAS
REFab	REFab $\xrightarrow{t_{RFC1}}$ NOP... (all banks blocked)
REFsb	REFsb(bank) $\xrightarrow{t_{RFCsb}}$ NOP (that bank only; others free)
ZQCS	MPC ZQCAL Start $\xrightarrow{t_{ZQCAL}}$ MPC ZQCAL Latch $\xrightarrow{t_{ZQLAT}}$ ready
MRW	MRW $\xrightarrow{t_{MRD}}$ next command

Output Signal Format

Field	Width	Description
cmd_type	3b	ACT / RD / WR / PRE / REF / NOP / MRS / ZQCS
rank	2b	Target rank
bank_group	3b	BG[2:0] for x4/x8; BG[1:0] for x16
bank	2b	Bank within BG
row	17b	Row address (used in ACT)
col	11b	Column address (used in CAS)
ap	1b	Auto-precharge flag (CA10)
wr_partial	1b	WR_PARTIAL: triggers internal RMW path in DRAM

Maintenance Engine

The Maintenance Engine runs four independent sub-FSMs that handle all autonomous background operations. Each posts requests to the Cmd Scheduler via a priority request bus.

Refresh Scheduler FSM

States: IDLE → REF_DUE → WAIT_BANKS_IDLE → ISSUE_REF → WAIT_tRFC → DONE → IDLE.

REFsb path (when REF_MODE=01): REF_DUE → WAIT_TARGET_BANK_IDLE → ISSUE_REFsb → WAIT_tRFCsb → DONE → IDLE.

- **Leaky-bucket counter:** increments by 1 each tREFI. On REF issued: decrement by 1. If counter reaches 9: **ref_urgent** asserted, hard-stalling all RD/WR.
- **REFsb cycling:** bank address cycles BA[1:0] in order 00→01→10→11. 8 REFsb commands ≈ 1 REFab credit.
- **FGR mode:** when REF_MODE[1]=1, tRFC used is tRFC2; when REF_MODE[1:0]=11, tRFC4. Budget counter threshold is halved or quartered accordingly.
- **Temperature derating:** if MR4 TUF=1, tREFI is halved before loading into the credit tracker. Controller polls MR4 periodically via background MRR.

ZQcal FSM

States: IDLE → WAIT_IDLE → ISSUE_ZQCAL_START → WAIT_tZQCAL → ISSUE_ZQCAL_LATCH → WAIT_tZQLAT → DONE → IDLE.

Triggered by: periodic tZQCS interval timer (CSR-configurable), or host write to ZQCAL_TRIG bit. All banks must be idle before issuing. During ZQCAL, all commands are blocked via gate_zqcs.

RFM FSM (Rowhammer Management)

States: IDLE → MONITOR_RAA → RFM_REQUEST → WAIT_ISSUE → WAIT_tRFM → UPDATE_RAA → MONITOR_RAA.

RAA counter logic:

- RAA[b] += 1 on every ACT to bank b .
- RAA[b] -= RAADec on every REF issued to bank b .
- When RAA[b] ≤ RAAIMT: assert RFM request to Scheduler (priority: above ZQcal, below refresh override).
- tRFM = tRFC1 (same duration). Bank enters RFM_ACTIVE during this window.
- After tRFM: RAA[b] reset to RAAIMT. Return to MONITOR_RAA.

Power Management FSM

States: NORMAL → PD_ENTRY_CHECK → PRECHARGE_PD or ACTIVE_PD → PDX_WAIT_tXP → NORMAL.

Self-Refresh branch: NORMAL → SR_ENTRY → WAIT_tCKSRE → SELF_REFRESHING → SR_EXIT → WAIT_tCKSRX → WAIT_tXS → WAIT_tDLLK → NORMAL.

- **Precharge PD:** entered from IDLE. All banks idle. CKE deasserted.
- **Active PD:** entered from BANK_ACTIVE. Bank retains open row. On PDX: bank resumes from BANK_ACTIVE after tXP.
- **Self-Refresh:** all banks must be precharged first. tCKSRE stable clock enforced after CKE falls. On SRX: tCKSRX stable clock, then tXS, then tDLLK for commands requiring locked DLL.
- **MPSM:** MR2 OP[1:0]=01. Deeper power state than SR. All banks precharged prerequisite. Exit via DES then MRW to clear MPSM.

Write Data Path

Sub-Block	Description
Write Data Buffer	BL16-aligned FIFO. Stores write data arriving from the CIF Write Cmd Pool via the CDC FIFO. Dequeued when the Cmd Scheduler confirms a write slot.
DM/ECC Generator	Computes per-byte Data Mask. Generates inline Write CRC using CRC-8 polynomial $G(x) = x^4 + x + 1$ over DQ and DM bits per BL16 burst. Write CRC enabled via MR8 OP[6]=1.
WL Align	Inserts a programmable pipeline delay equal to CWL nCK to align the DQS preamble to the DFI dfi_wrddata_en assertion. Offset loaded from PHY_WRLAT config reg.

Sub-Block	Description
DFI Write Timing	Drives <code>dfi_wrdata[127:0]</code> , <code>dfi_wrdata_mask[15:0]</code> , <code>dfi_wrdata_cs_n</code> , and <code>dfi_parity_in</code> . Write preamble per MR8 (2tCK default, 4tCK optional). ODT timing via tODTon = WL−2 and tODToff = WL+BL/2.
Write CRC Error FSM	Monitors <code>dfi_alert_n</code> from PHY. States: NORMAL → CRC_ERROR_DETECTED → RETRY_WRITE → AUTO_DISABLE . Retries up to n times (threshold from MR51/52). If retry count exceeded: auto-disables Write CRC via MRW to MR8, flags error to CIF error register.

Read Data Path

Sub-Block	Description
Read Latency Counter	Per-outstanding-read counter. Tracks CL + read preamble cycles from the RD command issue to expected data arrival on DFI. Issues <code>dfi_rddata_en</code> exactly PHY_RDLAT cycles before the expected <code>dfi_rddata_valid</code> .
Read Data FIFO	Captures <code>dfi_rddata[127:0]</code> when <code>dfi_rddata_valid</code> asserts. Re-aligns per-burst data and feeds the CIF Reorder Buffer. Tags read data with the AXI transaction ID for out-of-order return.
ECC/CRC Checker	Verifies Read CRC when MR8 OP[7]=1. On CRC error: flags to CIF error register, optionally triggers read retry. Read CRC adds a latency adder to RL (Table 365 of JESD79-5).
ECS FSM	States: IDLE → ECS_ACTIVE → WAIT_tECS → IDLE . Triggered by MR14 (auto-ECS mode) or host MPC request. DRAM scrubs its on-die ECC array during tECS window ($\leq$ tRFC1). Error logs readable from MR16–MR20.

DFI 5.2 Interface

DFI timing: `dfi_wrdata_en` must be asserted **PHY_WRLAT** cycles before write data. `dfi_rddata_valid` arrives **PHY_RDLAT** cycles after the RD command. Both offsets are PHY-specific and set during DFI training. Frequency ratio (**FREQ_RATIO**) determines how many nCK per DFI clock.

Signal	Direction	Description
<code>dfi_address[13:0]</code>	MC→PHY	CA bus. 2-cycle encoding for DDR5 ACT.
<code>dfi_cs_n[1:0]</code>	MC→PHY	Chip select per rank.
<code>dfi_cke[1:0]</code>	MC→PHY	Clock Enable per rank.
<code>dfi_odt[1:0]</code>	MC→PHY	On-Die Termination control per rank.
<code>dfi_reset_n</code>	MC→PHY	DRAM RESET_n driven by Init FSM.
<code>dfi_act_n</code>	MC→PHY	Activate indicator (DDR5).
<code>dfi_bank[1:0]</code>	MC→PHY	Bank address.
<code>dfi_bg[2:0]</code>	MC→PHY	Bank group (BG[2:0] x4/x8; BG[1:0] x16).
<code>dfi_wrdata[127:0]</code>	MC→PHY	Write data bus (BL16 × DQ8 per sub-channel).
<code>dfi_wrdata_mask[15:0]</code>	MC→PHY	Write data mask per byte lane.

Signal	Dir	Description
dfi_wrdata_en	MC→PHY	Write data enable. Asserted PHY_WRLAT cycles early.
dfi_wrdata_cs_n	MC→PHY	Per-DRAM write chip select (DDR5 PDA).
dfi_parity_in	MC→PHY	CA parity bit (DDR5).
dfi_rddata[127:0]	PHY→MC	Captured read data from PHY.
dfi_rddata_valid	PHY→MC	Read data valid. Read Data FIFO captures on this.
dfi_rddata_dnv	PHY→MC	Data-not-valid error flag.
dfi_alert_n	PHY→MC	DRAM alert: Write CRC error or CA parity error.
dfi_ctrlupd_req	MC→PHY	Controller requests PHY update (periodic).
dfi_ctrlupd_ack	PHY→MC	PHY acknowledges update request.
dfi_phyupd_req	PHY→MC	PHY requests MC to pause command issue.
dfi_phyupd_ack	MC→PHY	MC acknowledges PHY update pause.
dfi_init_start	MC→PHY	Init FSM: start PHY initialization handshake.
dfi_init_complete	PHY→MC	PHY initialization done.
dfi_freq_ratio[1:0]	MC→PHY	Frequency ratio: 00=1:1, 01=1:2, 10=1:4.

FSM Inventory

FSM	States	Instances	Notes
Init FSM	16	1	Longest chain; RESET_n through training complete
Bank State Machine	11	16	One per bank; 176 total state bits
Arbitration Logic	6	1	RD/WR mode + priority selection
Command Selector	4	1	Page policy + legal check + sequence gen
Refresh Scheduler	7	1	REFab + REFsb + FGR + urgent override
ZQcal FSM	7	1	MPC ZQCAL Start/Latch flow
RFM FSM	6	1	Per-bank RAA monitoring + RFM injection
Power Management FSM	10	1	PD (Pre/Act), SR, MPSM
Write CRC Error FSM	4	1	dfi_alert_n monitoring; retry + auto-disable
ECS FSM	4	1	DDR5 inline ECC scrub window
Total	≈75	23	

Timing Reference

All values at @200 MHz (tCK=5 ns, CL=3, CWL=1, BL/2=8) and @100 MHz (tCK=10 ns). These are the simulation-corner reference values used during design verification. Production values are set via Config Regs based on actual speed bin.

Command Sequence	Formula	@200	@100
ACT → RD/WR (same bank)	tRCD	4	4
ACT → PRE (same bank, min)	tRAS	7	4
PRE → ACT (same bank)	tRP	4	4
ACT → ACT (same bank)	tRC = tRAS+tRP	11	8
ACT → ACT (same BG)	tRRD_L	4	4
ACT → ACT (diff BG)	tRRD_S	2	2
RD → RD (same BG)	tCCD_L	8	8
WR → WR (same BG)	tCCD_L_WR	32	32
WR → WR same BG (not RMW)	tCCD_L_WR2	16	16
RD/WR (diff BG)	tCCD_S = BL/2	8	8
RD → WR (any)	RL+BL/2-WL+2+tWPRE	14	14
WR → RD (same BG)	WL+BL/2+tWTR_L	13	13
WR → RD (diff BG)	WL+BL/2+tWTR_S	11	11
RD → PRE (same bank)	tRTP	4	4
WR → PRE (same bank)	CWL+BL/2+tWR	15	12
WRA → ACT (same bank)	CWL+BL/2+tWR+tRP	19	16
RDA → ACT (same bank)	tRTP+tRP	8	8
REFab → any	tRFC1	39	20
SRX → ACT (no DLL)	tXS = tRFC1+10 ns	41	21
PDX → any	tXP	5	5
MRW → MRW	tMRD	16	16
ZQCAL Start → Latch	tZQCAL = 1 ps	200	100
ZQCAL Latch → next cmd	tZQLAT	30	30
4 ACT window	tFAW	16	16

Constraint Support Checklist

Constraint	Status	Notes
CAS-to-CAS		
tCCD_L same BG RD→RD		max(8 nCK, 5 ns); BG tracker
tCCD_S diff BG RD→RD		8 nCK
tCCD_L_WR same BG WR→WR		max(32 nCK, 20 ns)
tCCD_L_WR2 same BG WR→WR (2nd not RMW)		max(16 nCK, 10 ns)
tCCD_S_WR diff BG WR→WR		= tCCD_S = 8 nCK
ACT-to-ACT		
tRRD_L same BG		max(8 nCK, 5 ns)
tRRD_S diff BG		8 nCK
tRRD_L(1K) / tRRD_L(2K)		1 KB and 2 KB page size differentiation
tFAW 4-ACT window		4-entry shift register of ACT timestamps
ACT / PRE / Row Timing		
tRCD, tRP, tRAS		Per-bank countdown timers

Constraint	Status	Notes
tRC = tRAS+tRP		Implicit from tRAS + tRP timers
tRTP, tWR		Per-bank timers
tRAS max (row open timeout)	x	Force-PRE on 9×tREFI not implemented
Write/Read Turnaround		
tWTR_L same BG		WL+BL/2+tWTR_L
tWTR_S diff BG		WL+BL/2+tWTR_S
tRTW		RL+BL/2−WL+2
Refresh		
tRFC1 REFab recovery		Per-rank block via gate_rfc
tRFCsb REFSb per-bank		Only target bank blocked
tREFI tracking		Leaky-bucket; up to 8 postponements
ref_urgent hard stall		Asserted at credit = 8
tREFI derating (>85°C)		2× via MR4 TUF polling
Per-bank REFSb credit balance	x	Global credit only; per-bank balance not tracked
Automatic BA cycling REFSb	x	BA rotation not enforced in hardware
FGR 4× mode	x	FGR 2× supported; 4× not
ZQ Calibration		
tZQCAL, tZQLAT		MPC ZQCAL Start/Latch
Periodic tZQCS		CSR-configurable interval
Mode Registers		
tMRD (MRW→MRW)		
tRFC1 ODT: tODTon/tODToff		WL−2 / WL+BL/2
Dynamic RTT switching	x	Static RTT programming only
DFI Latency		
dfi_t_rddata_en		RL cycles ahead of CAS
dfi_t_phy_wrlat		WL cycles ahead of data
dfi_freq_ratio 1:1/1:2/1:4		
Read CRC RL adder	x	Not pipelined
Power / Self-Refresh		
tCKSRX, tXS		Init FSM + Power Mgmt FSM
MPSM entry/exit	x	All-banks-precharged prerequisite not enforced
Power-down entry/exit	x	Dynamic power-down not supported
RFM / Rowhammer		
RAA counter per bank		RFM FSM tracks per-bank RAA
RAAIMT threshold monitoring		From MR58 mirrored to config reg
RFMab injection		Via Maintenance Engine priority request
RFMSb injection	x	Not implemented
ALERT_n RFM decode	x	
Multi-Rank		
Per-rank tRFC isolation		gate_rfc[r] per rank
tRRD across ranks		
Rank interleave during tRFC		Other rank serviced during tRFC bubble
Not Supported / Deferred		
tRASmax / force-PRE timeout	x	

Constraint	Status	Notes
BL32 timing adjustments	x	
2N mode CA pipeline	x	
PPR sequence	x	Guard key + hPPR/sPPR flow
ECS scrub address walking	x	MR14–MR21 scrub scheduling
DQS oscillator scheduling	x	Post-init re-cal trigger
CA parity retry	x	ALERT_n decode + retransmit
Write Pattern command (WRP)	x	

Supported: 51 Not Supported / Deferred: 27 Total: 78

DDR5 vs DDR4 — Design Deltas

Feature	DDR4	DDR5 — MC Core Impact
BL	BL8 (BL/2=4)	BL16 (BL/2=8). All RTW, WTR, WR-to-PRE formulae scale by 2.
ACT encoding	1-cycle CA	2-cycle CA. Scheduler must drive CA on two consecutive rising edges.
REFsb	Not supported	Refresh Scheduler adds REFsb path; per-bank tRFCsb timer in BSM.
RFM	Not supported	Full RFM FSM; per-bank RAA counters; RAAIMT/RAAMMT in Config Regs.
ZQ mechanism	ZQCL/ZQCS commands	MPC ZQCAL Start+Latch. Init FSM restructured; ZQcal FSM uses MPC.
tRFC1 (8 Gb)	350 ns	195 ns. Shorter penalty; more scheduling headroom.
tREFI	7.8 μ s	3.9 μ s. Double refresh rate; budget counter threshold halved.
CWL	Separate lookup table	CWL = CL–2 (fixed). Simplifies Config Reg and RTW formula.
tMOD	24 nCK after any MRS	Replaced by tMRD only (16 nCK). No separate tMOD path needed in Scheduler.
tCCD_L_WR	Not defined	max(32 nCK, 20 ns). New gate in Command Selector.
Write CRC	Optional (MR5)	Mandatory architecture support. dfi_alert_n monitoring; Write CRC Error FSM required.
ECS	Not supported	ECS FSM in Read Data Path; hooks into tRFC window; MR14–MR20 support.
MR addressing	3 MRs (MR0/1/2/3)	256 MRs (MR0–MR255) via MRW/MRR commands. Init FSM MRW burst extended.